DTU

Model-based Machine Learning

Project description

Outline

Rules

Important dates

Common mistakes, misconceptions and advice

Take-away ideas

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  - **Model-driven:** take a few modeling ideas covered during the lectures and put them together to create a new model, implement it in Pyro and test it
  - **Paper-driven:** find a paper with a cool probabilistic modeling approach, formulate it as a PGM+generative process, implement it in Pyro and test it
- If in doubt, talk with us!
- We also provide some suggestions for projects

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- 15 May - Final delivery (in groups)
  - **Fully self-explanatory** notebook (like the ones we provide in class)
  - 6-page report (incl. figures and tables) double-column IEEE template - **research paper style**

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  - Consider for example: Intelligence $\rightarrow$ Course grade $\rightarrow$ Recommendation letter
  - If course grade is always observed, then it blocks the path between intelligence and recommendation letter. Therefore, "Intelligence $\rightarrow$ Course grade" and "Course grade $\rightarrow$ Recommendation letter" become two independent models.
  - There is no point estimating them jointly because there is no flow of information between the two (unless there is an alternative path, of course. . .)

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  - There is no point estimating them jointly because there is no flow of information between the two (unless there is an alternative path, of course. . .)
- Don't just think about what variables depend on other variables, and their distribution types. Think also of how you should model those dependencies (e.g. how to condition the parameters of a Beta distribution on another variable?)

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- Use ancestral sampling to generate artificial data, and run inference on the model using Pyro to see if it is able to recover the true values/parameters that were used to generate the data. This is a great way of guarantying that the model is correctly implemented, and that inference is working correctly

Analysis group membership in Social Networks I

- Graph data of human interaction is called social networks. The interaction could for example being friends on Facebook or talking to each other.
- The project inferred groups in social networks based on Game Of Thrones. Take a look at the data here https://github.com/mathbeveridge/gameofthrones
- Tried different models. Results below is from a Mixed membership stochastic block model - MMSBM.

Analysis group membership in Social Networks II

1. For $n = 1..N$
   1. draw $\boldsymbol{\pi}_n \sim \text{Dirichlet}(\boldsymbol{\alpha})$
2. For $k, k' = \{1, 2, \ldots, K\} \times \{1, 2, \ldots, K\}$ where $k \neq k'$
   1. draw $\lambda_{k,k'} \sim \text{Gamma}(a, b)$
3. For k in $1..K$
   1. draw $\lambda_{k,k} \sim \text{Gamma}(c, d)$
4. For all pairs $i, j \in \{1, 2, ...N\} \times \{1, 2, ...N\}$ where $i \neq j$
   1. $x_{i,j} \sim \text{Poisson}(\boldsymbol{\pi}_i^T \lambda \boldsymbol{\pi}_j)$;

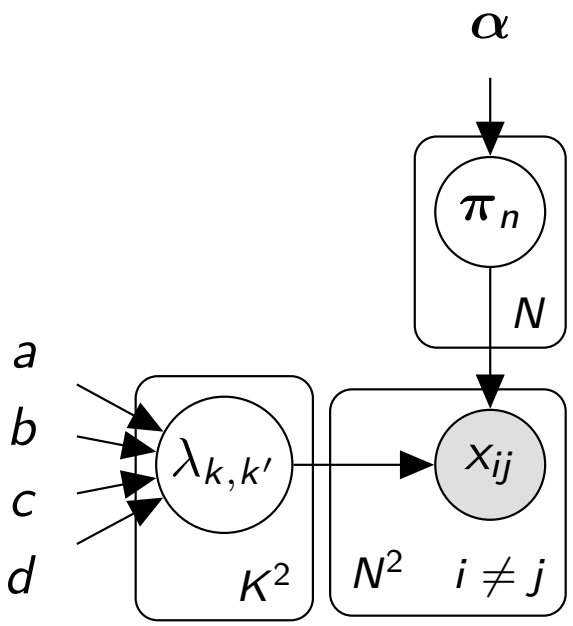


Figure: PGM for Mixed membership stochastic block model

Analysis group membership in Social Networks III

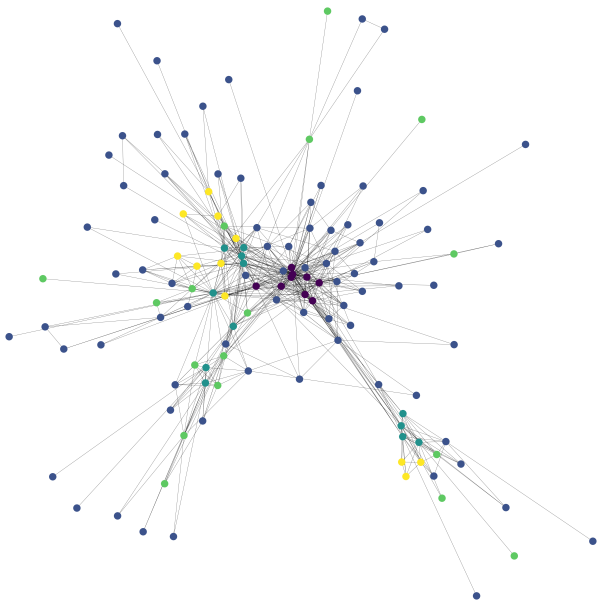


Figure: Each node is a GoT character from season 5. Colors indicate one of 5 inferred groups.

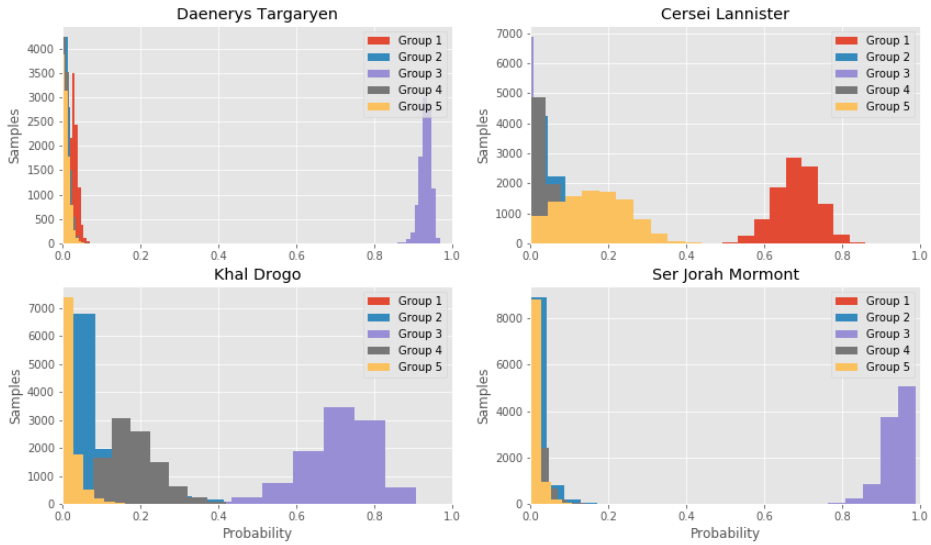


Figure: The histogram shows posterior probabilities of belonging to one of 5 groups for 4 main characters of GoT

Bayesian semi-parametric soccer analysis

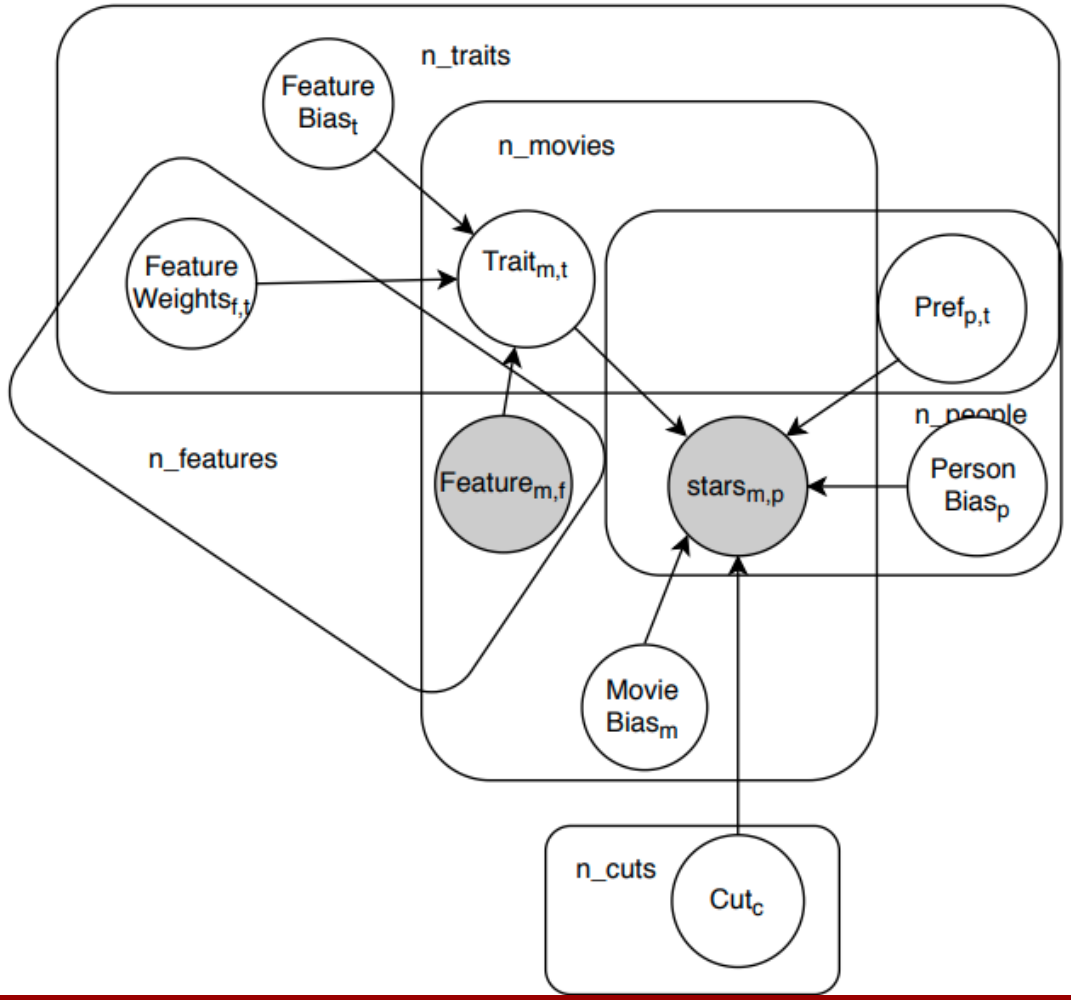
- Apply the time-to-event analysis methods to analyze the intensity with which goal scoring occurs in soccer matches
- Tried different time-to-event models from the class of Cox proportional hazards models and extended these to incorporate non-linear effects by use of the highly flexible Gaussian processes from the Bayesian non-parametric toolbox
- The hazard in the i-th match at time interval j is

$$\lambda_{ij} = \lambda_j \exp(\beta_1 * \mathit{skill_gap} + \beta_2(t) * \mathit{x_goal})$$

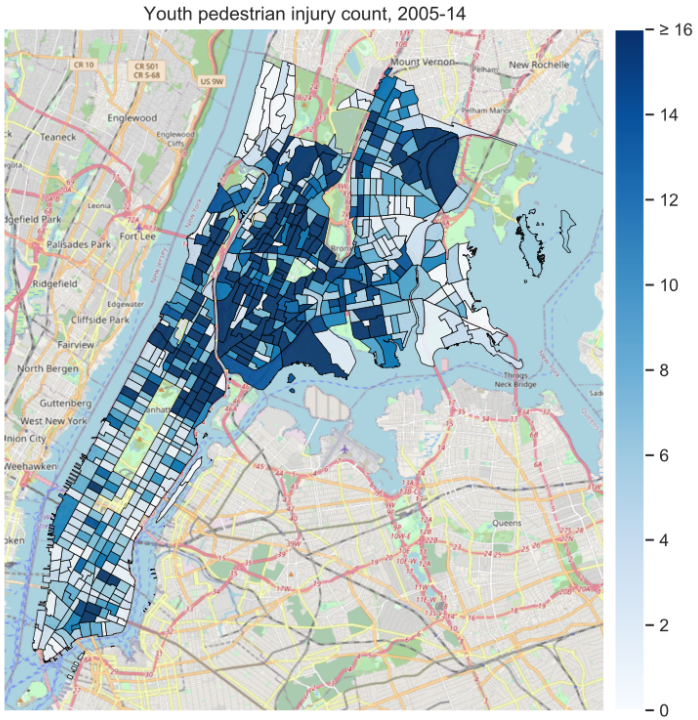
- Baseline of Cox model is a GP: $\log \lambda_0 \sim \mathcal{GP}(\mathbf{0}, K(t, t'))$

Movie recommendations - MovieLens dataset

- Inspired by the example in Bishop's "Model-based Machine Learning" book, but with a good mix of creativity, lots of experimentation and insightful discussions



Bayesian Spatial Count Models



- For each area i, the observed count y_i follows a Negative Binomial distribution

$$y_i \sim \text{NB}(r, p_i), \qquad \text{(likelihood)}$$

$$p_i = \frac{\exp(\psi_i)}{1 + \psi_i}, \qquad \text{(link function)}$$

$$\psi = \beta^T \mathbf{X}_i + \phi_i, \qquad \text{(linear function of inputs } \mathbf{X}_i\text{)}$$

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- **Research topic 1**: How to have non-linear dependencies on inputs $\mathbf{X}_i$?
  - Neural networks?
  - Hierarchical modelling?

[1] Reference: `https://arxiv.org/abs/2007.03681`

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- **Research topic 1**: How to have non-linear dependencies on inputs $\mathbf{X}_i$?
  - Neural networks?
  - Hierarchical modelling?
- **Research topic 2**: How to capture spatial correlations between areas?
  - Capture correlations through the noise term?[1]

$$\exp(\tau \mathbf{W})\phi \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$$

  - Neural networks? E.g., graph neural nets?

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Modelling traffic accident injury severity levels I

- Traffic accident reports typically contain information about the severity of the injury sustained by each individual involved in the accident
- Injury severity is reported on an ordinal scale with levels “no injury”, “light injury”, “severe injury”, “fatal injury”

Modelling traffic accident injury severity levels II

- Modelling injury severity
  - Disaggregate $\rightarrow$ let y_i denote the injury severity sustained by individual i.
    - Categorical distribution

$$y_i \sim \text{Categorical}(y_i|\text{Softmax}(X_i, \theta)) \tag{1}$$

    - Ordered logit

$$\begin{aligned} P(y_i = s) &= P(\kappa_{s-1} < f(X_i, \theta) < \kappa_s) \\ &= \frac{1}{1 + \exp(-\kappa_s + f(X_i, \theta))} - \frac{1}{1 + \exp(-\kappa_{s-1} + f(X_i, \theta))}, \end{aligned} \tag{2}$$

      where κ_s with $s = 1, \ldots, S$ are estimable cut-off parameters. Be sure to carefully check the exact parametric for the identification restrictions.
  - Aggregate $\rightarrow$ aggregate accidents over space and time by severity level
    - Multivariate count data

Modelling traffic accident injury severity levels III

- Dataset examples
  - https://discover.data.vic.gov.au/dataset/crash-stats-data-extract
  - https://opendata-nzta.opendata.arcgis.com/search?collection=Dataset&tags=CAS
- **Research ideas:**
  - Focus on the specific type of accident (e.g., crashes involving pedestrians, cyclists, trucks, exactly two vehicles, more than two vehicles, etc.)
  - Exploit hierarchical structure of data $\rightarrow$ Individuals are nested within vehicles; vehicles are nested within accident events; accident events are nested within time periods, ...
  - Account for reporting errors
  - Extend to non-linear functions (e.g., neural nets or Gaussian processes)
  - Analyse temporal evolution of determinants of injury severities
  - ...